# Tattoo inks (permanent and temporary): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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## Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for tattoo inks (permanent and temporary) in Cosmetics and Personal Care: Leave-on, evaluated on finished leave-on cosmetic, TOTAL CONTENT as placed on market (ug/kg of product as sold), on the technically-unavoidable--impurity basis every instrument in this family uses. Not a migration/leachate basis, not a dietary basis, not an ingredient purity specification.. 6 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

## 1 | How every number on this standard is set

HMTc Method v2.0, government-limit-only initial setting: every numeric cell is the strictest applicable sovereign maximum for this exact leave-on cosmetic form expressed as total content of the finished product as placed on market, or a disclosed read-across where the exact cell is unregulated, or a substantiated not-material determination, or a control. No cell is occurrence-derived, no cell is precautionary or feasibility-adjusted, and no cell is anchored on an ingredient or colour-additive purity specification.

## 2 | Master limit table

All values in ppb (µg/kg), as_consumed basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 700 | government maximum level |

| | | | |
|---|---|---|---|
| Cadmium (Cd) | Tier 1 | 100 | government maximum level |
| Total arsenic (tAs) | Tier 2 | 500 | government maximum level |
| Inorganic arsenic (iAs) | Tier 1 | ≤ tAs | bounded by tAs (reflex speciation) |
| Mercury, total (tHg) | Tier 1 | 100 | government maximum level |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 5,000 | government maximum level |
| Tin (Sn) | Tier 2 | control | measured every lot; no human-food maximum |
| Aluminium (Al) | Tier 2 | control | measured every lot; no human-food maximum |
| Total chromium (Cr) | Tier 2 | control | measured every lot; no human-food maximum |
| Hexavalent chromium (Cr-VI) | Tier 2 | 500 | government maximum level |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 700 | government maximum level |
| Cadmium (Cd) | 100 | government maximum level |
| Total arsenic (tAs) | 500 | government maximum level |
| Inorganic arsenic (iAs) | ≤ tAs | iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Mercury, total (tHg) | 100 | government maximum level |

| | | |
|---|---|---|
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 5,000 | government maximum level |
| Tin (Sn) | control | The register carries no tin row for the EU tattoo and permanent-make-up entry, and no jurisdiction publishes a tin maximum for a finished cosmetic either, so neither instrument governing this row supplies a tin number. Whether Appendix 13 sets a tin concentration limit for these mixtures has NOT been verified against the instrument, and the register carries no tin row for the entry; this standard does not fill a cell from recollection, so the question is recorded as a register gap for resolution through the verified regulatory-fetch pathway and the cell is a control in the meantime. If a sovereign tin value is confirmed and ingested, this cell is re-derived as a numeric cell. The register's other tin values are dietary maximum levels and food-contact or skin-contact organotin limits, in bases that cannot anchor a total-content mixture cell. Tin is a real presence here through tin-oxide pigments and organotin stabiliser residues in ink carriers. |
| Aluminium (Al) | control | Neither instrument governing this row sets an aluminium maximum: the EU tattoo and permanent-make-up entry lists no aluminium value, and no jurisdiction publishes an aluminium maximum for a finished cosmetic. There is no near-neighbour cell to read across from. Aluminium is nonetheless intrinsic to this product form rather than incidental, because aluminium-lake pigments and alumina substrates are standard ink chemistry, so a single contaminant ceiling would be meaningless while the metal cannot go unmeasured. |

| | | |
|---|---|---|
| Total chromium (Cr) | control | Neither instrument governing this row sets a total-chromium maximum. The EU tattoo and permanent-make-up entry regulates chromium only as Cr(VI), which is the deliberate and correct choice: total chromium in an ink is dominated by benign trivalent chromium from chromium-oxide and chromium-hydroxide green pigments, so a total-chromium ceiling would not be protective as a standalone number. No jurisdiction publishes a total-chromium maximum for a cosmetic either. Total chromium is therefore carried as the mandatory screen that governs the Cr(VI) reflex on this row, and no ceiling is invented. |
| Hexavalent chromium (Cr-VI) | 500 | government maximum level |

## 4 | Cite this standard

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### STATUS AND LICENCE

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### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://-heavymetalcertified.com/standards/tattoo-inks. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

### PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/tattoo-inks.